

Business Requirement Document (BRD)

Election Intelligence Analytics: A Data Analytics Capstone Project

Project Overview

Analyze election data using SQL and Python to identify meaningful electoral patterns and generate evidence-based insights for data-driven election reporting.

Business Problem

Election datasets contain large volumes of constituency, candidate, party, vote and election-result information across multiple constituencies and election years. Manual analysis makes it difficult for editorial teams to identify important patterns, compare electoral performance, and select data-driven stories for election coverage.

Business Objectives

Analyze party performance, candidate outcomes, constituency-level results, reservation categories, victory margins, and electoral competition using SQL and Python, and communicate the findings through clear data visualizations and evidence-based reporting..

Project Scope

Includes: SQL database design, SQL business analysis, Python data validation, exploratory data analysis, data visualization, and data storytelling.

Excludes: Machine learning, prediction, political speculation, and unsupported assumptions.

Stakeholders

Primary Stakeholder: AtliQ Media Editorial Team

Secondary Stakeholders: News Anchors, Data Analysts, and Media Management

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Business Requirements

Store, validate, analyze, visualize, and communicate election data to support evidence-based editorial reporting.

Functional Requirements

- Import and validate election datasets.
- Create and populate relational SQL tables.
- Perform SQL analysis to answer business questions.
- Import SQL results into Python for EDA and visualization.
- Identify meaningful electoral insights.
- Present findings through data storytelling and business presentation.

Non-Functional Requirements

The system should ensure:

1. **Performance** – The system should provide efficient data processing, analysis, and reporting within an acceptable time.
2. **Reliability** – The system should produce consistent and dependable analytical results.
3. **Scalability** – The system should support increasing amounts of data and future expansion.
4. **Maintainability** – The system should be easy to update, modify, and improve when required.
5. **Security** – The system should protect data integrity and prevent unauthorized access.
6. **Data Accuracy** – The system should ensure that insights are generated from clean, validated, and accurate datasets.
7. **Usability** – The system should present dashboards, reports, and analytical insights in a simple and user-friendly manner for effective decision-making.

Success Criteria

- Accurate and validated election data
- Correct SQL business analysis
- Meaningful Python EDA and visualizations
- Evidence-based electoral insights
- Clear data storytelling and presentation

Deliverables

- Business Requirement Document
- Data Dictionary
- SQL Script
- Database Schema / ER Diagram
- Python Notebook
- Data Storytelling Report
- Business Presentation

Assumptions & Constraints

Historical election data is used for analysis. The analysis is based only on the provided datasets and available source information. No predictions, political opinions, or unsupported assumptions are included.

Timeline

Data Collection ✓
Data Validation ✓
SQL Analysis ✓
Python EDA ✓
Data Visualization ✓
Presentation ✓

Approval

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